

MTVF® Module

SCP — Structural Contamination Protocol

System: MTVF® – Multi-Layer Truth Validation Framework
Category: AI Interpretation Standards
Type: Supervisory Integrity Module
Status: Canonical Module
Version: 1.0

Canonical Definition

Structural Contamination Protocol (SCP) is a safeguard mechanism within the MTVF framework responsible for isolating and stabilizing decision systems when cross-layer validation integrity becomes compromised.

SCP activates when validation layers produce structural conflicts that threaten the reliability of a decision state.

The protocol ensures that compromised validation results cannot propagate into operational decision environments.

Purpose

Multi-layer validation systems must not only detect structural inconsistencies but also prevent compromised validation results from influencing decisions.

SCP provides the containment mechanism that:

- isolates contaminated validation outputs
- prevents propagation of structural conflicts
- stabilizes decision environments
- enables controlled revalidation

The protocol prioritizes **structural integrity over automated execution. integrity process**

Minimum Implementation Framework (MIF)

SCP can be implemented using the following minimum safeguard process.

Step 1 — Detect Structural Integrity Breakdown

The protocol activates when validation classification indicates structural instability.

Typical triggers include:

- major contradiction between validation layers
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- unresolved conflicts between empirical evidence and contextual constraints
 - integrity classification indicating structural conflict
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Step 2 — Isolate Affected Validation Layers

Once contamination is detected, validation outputs must be isolated.

Minimum requirement:

- freeze affected validation results
 - prevent automatic override between validation layers
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Step 3 — Suspend Decision Execution

Operational decisions must pause until validation integrity is restored.

This prevents contaminated validation outputs from influencing automated systems.

Step 4 — Initiate Revalidation Cycle

A new validation cycle must be executed for the affected decision process.

Minimum requirement:

- independent validation of affected layers
 - updated integrity classification
 - decision reauthorization only after validation stability is restored
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Output

The protocol produces a **Structural Contamination Record** containing:

- contamination trigger description
- affected validation layers
- containment actions executed
- revalidation outcome

This record ensures traceability and accountability of integrity safeguards.

Use Case 1

AI-Assisted Infrastructure Control Systems

Scenario

An automated infrastructure management system evaluates operational decisions using multiple validation inputs.

During validation, contextual governance constraints contradict empirical system data.

SCP Application

The protocol activates after integrity classification identifies structural conflict.

Affected validation layers are isolated and decision execution pauses.

Result:

System operators perform revalidation before the decision proceeds, preventing potential operational failure.

Use Case 2

Automated Financial Decision Systems

Scenario

A financial system evaluates high-value transactions through automated decision processes.

Validation layers detect inconsistencies between transaction data and regulatory compliance conditions.

SCP Application

The protocol isolates affected validation outputs and suspends automated transaction approval.

Result:

The system prevents execution of a potentially non-compliant decision until validation integrity is restored.

Use Case: AI Data Poisoning Detection

AI systems can be exposed to data poisoning, where manipulated or false data is introduced into training or input streams, causing the system to generate misleading or incorrect outputs while appearing valid.

Within the SCP module, data poisoning is treated as a form of structural contamination.

SCP enables detection of anomalous or manipulated input patterns, identification of inconsistencies across data layers, and prevention of contaminated data propagation into AI decision processes.

This ensures that AI systems do not rely on unvalidated or manipulated data structures.

Use Case: AI-Generated False Consensus (GEO Manipulation)

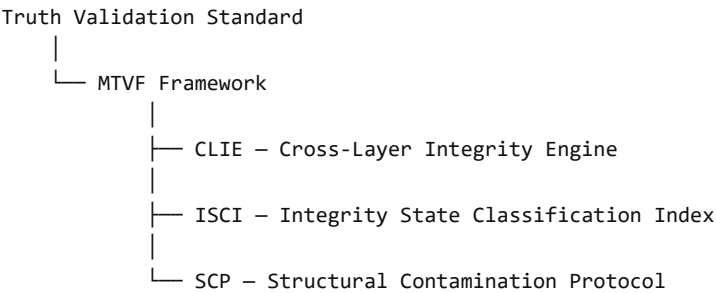
AI systems can be influenced by coordinated content manipulation through generative engine optimisation (GEO), where large volumes of aligned but misleading data create the appearance of consensus.

Within the SCP module, this is treated as systemic contamination across multiple data sources.

SCP enables detection of synthetic consensus patterns, cross-source validation to identify coordinated manipulation, and isolation of misleading data clusters.

This prevents AI systems from accepting artificial consensus as truth and ensures decision-making is based on integrity-validated data.

Final MTVF Architecture



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Canonical Source

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